

Vishwajeet Kumar

Software Engineer | Backend & AI Systems

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EDUCATION

Bachelor of Technology in Computer Science (Artificial Intelligence)

Nov 2022 – Jun 2026

University of Lucknow

CGPA: 8.16 / 10

SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL

Backend & AI Frameworks: FastAPI, Spring Boot, NestJS, Node.js, Express.js, LangChain, LangGraph, REST APIs

Databases & Caching: PostgreSQL, MongoDB, Redis

Cloud & DevOps: AWS (EC2, RDS, S3), Docker, Kubernetes, GitHub Actions, CI/CD, Git

Core CS & Practice: Data Structures & Algorithms (500+ LeetCode), System Design, Distributed Systems, DBMS, OOP

OPEN SOURCE CONTRIBUTIONS

LangChain — Open Source Contributor (142k+ GitHub Stars)

- Engineered comprehensive input validation and case-insensitive model inference for the core `init_chat_model` function, adding explicit error handling while maintaining 100% backward compatibility.
- Implemented 6 unit test suites with parametrized coverage; successfully merged PR after passing 47 CI/CD checks and peer code reviews with core maintainers.

EXPERIENCE

Software Development Engineer Intern

Mar 2026 – Jun 2026

PurpleMerit

- Engineered production backend microservices for referral, wallet, and payout workflows using FastAPI, MongoDB, and JWT-based authentication, ensuring secure and reliable transaction handling.
- Optimized MongoDB aggregation pipelines and indexing strategies, reducing query latencies and supporting high-throughput payment processing under concurrent load.
- Containerized services with Docker and configured automated CI/CD deployment pipelines to AWS Kubernetes clusters via GitHub Actions for continuous delivery.

Software Engineer Intern

Jul 2025 – Sep 2025

StuFit Approach Pvt. Ltd.

- Designed a modular RESTful backend supporting Role-Based Access Control (RBAC), analytics, and user workflows for 2,000+ active users using NestJS and PostgreSQL.
- Improved database response times through relational query optimization, indexing, and normalized schema design.
- Authored unit tests with Jest to improve code reliability and documented API specs for seamless frontend integration.

PROJECTS

Distributed Job Scheduler | Java, Spring Boot, PostgreSQL, Redis, Docker, Kubernetes, AWS | *GitHub*

- Eliminated duplicate job execution across distributed workers using Redis SETNX locking with Lua-scripted atomic release, and resolved database polling race conditions via PostgreSQL SELECT FOR UPDATE SKIP LOCKED across 50-job batch polls.
- Built heartbeat-based crash detection achieving zero orphan-job loss on worker failures, automatically recovering in-flight jobs within 30 seconds via exponential backoff retry (5s → 30s → 120s).
- Designed a 3-tier Redis priority queue (BRPOP-based) preventing starvation of critical jobs across a bounded worker thread pool (10–20 threads, 100-job queue depth).

EduIntel — AI Career Intelligence Platform | Python, FastAPI, LangChain, React, MongoDB, Redis | *GitHub*

- Built a full-stack AI career platform with 96 FastAPI routes and 49 reusable React components spanning resume analysis, ATS optimization, peer benchmarking, and interview practice.
- Built a peer-benchmarking pipeline over 500 synthetic profiles using K-Means, DBSCAN, Isolation Forest, and PCA across 10 profile features to surface comparative career insights and detect outlier resumes.

Real-Time Multiplayer Backend System | Node.js, WebSockets, Redis, PostgreSQL, AWS | *GitHub*

- Built a real-time multiplayer backend supporting 250+ concurrent users, implementing matchmaking and state synchronization using WebSockets, Redis Pub/Sub, and PostgreSQL.
- Achieved 600+ events/sec throughput with sub-10ms latency via optimized in-memory state management and efficient event routing.

CERTIFICATIONS

- Samsung Innovation Campus — Artificial Intelligence
- Redis — Get Started with Redis (Redis University)